UNIT: III

Electromagnetic Waves & Waveguides

Waves in general: plane waves in lossless dielectric, free space, Good conductor, Rectangular waveguide: Transverse Magnetic (TM) and Transverse Electric (TE) mode, wave propagation in guide, power transmission & Attenuation.

* Waves are means of transporting energy or information
* Typical examples of EM waves
  - radio waves
  - TV signals
  - Radar beams
  - light rays.
* All forms of EM energy share three fundamental characteristics
  - They all travel at high velocity
  - in travelling, they assume the properties of waves
  - they radiate outward from a source, without benefit of any discernible physical vehicles.

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In this chapter, our major goal is to solve maxwell's equations and describe EM wave motion in the following media.

1) free space ($\sigma=0, \varepsilon=\varepsilon_0, \mu=\mu_0$)

2) lossless dielectrics ($\sigma=0, \varepsilon=\varepsilon_r\varepsilon_0, \mu=\mu_r\mu_0$, or $\sigma \ll \omega\varepsilon$)

3) Lossy dielectrics ($\sigma\neq 0, \varepsilon=\varepsilon_r\varepsilon_0, \mu=\mu_r\mu_0$)

4) Good conductors ($\sigma \approx \infty, \varepsilon=\varepsilon_0, \mu=\mu_r\mu_0$ or $\sigma \gg \omega\varepsilon$)

$\omega \rightarrow$ angular frequency of the wave.

Wave Propagation in lossy Dielectrics

A lossy dielectric is a medium in which an EM wave, as it propagates, loses power owing to imperfect dielectric.

In other words,

a lossy dielectric → partially conducting medium (imperfect dielectric or imperfect conductor) with $\sigma\neq 0$

lossless dielectric → perfect or good dielectric with $\sigma=0$.

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Consider a linear, isotropic, homogeneous, lossy dielectric medium that is charge free ($\rho_v = 0$)

Assuming and suppressing the time factor $e^{j\omega t}$, Maxwell's equations become

$\nabla \cdot D = \rho_v$ $\quad D = \varepsilon E$

$$\nabla \cdot E_s = 0 \quad \text{—(1)}$$

$\nabla \cdot E_s = 0$

$\nabla \cdot B = 0$

$$\nabla \cdot H_s = 0 \quad \text{—(2)}$$

$\nabla \times E = -\frac{\partial B}{\partial t} = -j\omega\mu H_s$ $\quad B = \mu H$

$$\nabla \times E_s = -j\omega\mu H_s \quad \text{—(3)}$$

$$\nabla \times H_s = (\sigma + j\omega\varepsilon) E_s \quad \text{—(4)}$$

$J = \sigma E$

$\nabla \times H = J + \frac{\partial D}{\partial t}$

$= \sigma E_s + j\omega\varepsilon E_s$

$= (\sigma + j\omega\varepsilon) E_s$

Taking the curl of both sides of eqn (3)

$$\nabla \times \nabla \times E_s = -j\omega\mu (\nabla \times H_s) \quad \text{—(5)}$$

Applying the vector identity

$$\nabla \times (\nabla \times A) = \nabla (\nabla \cdot A) - \nabla^2 A$$

Apply the vector identity to eqn (5) (only LHS) & sub (4) in RHS

$$\nabla (\nabla \cdot E_s) - \nabla^2 E_s = -j\omega\mu (\sigma + j\omega\varepsilon) E_s$$

$= 0$ (1)

$$-\nabla^2 E_s = -j\omega\mu (\sigma + j\omega\varepsilon) E_s$$

$$\nabla^2 E_s - \gamma^2 E_s = 0 \quad \text{—(6)}$$

where $\gamma^2 = j\omega\mu(\sigma + j\omega\varepsilon)$ —(7)

(8)

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γ in reciprocal meters is called the propagation constant of the medium.

Similarly for H-field

$$\nabla^2 H_s - \gamma^2 H_s = 0 \quad \text{——} (8)$$

equations (6) & (8) are known as homogeneous vector Helmholtz's equation or vector wave equation.

Since $\gamma \rightarrow$ is a complex quantity

let $\gamma = \alpha + j\beta$ —— (9) $\Rightarrow \gamma^2 = (\alpha+j\beta)^2$

$= \alpha^2 + 2j\alpha\beta + (j\beta)^2$

$\gamma^2 = \alpha^2 + 2j\alpha\beta - \beta^2$. —— (10)

w.k.t $\gamma^2 = j\omega\mu(\sigma + j\omega\varepsilon)$

$\gamma^2 = j\sigma\omega\mu - \omega^2\mu\varepsilon$ —— (11)

$\gamma^2 = \alpha^2 - \beta^2 + 2j\alpha\beta$

$\Downarrow$

$\omega^2\mu\varepsilon$

Comparing (10) & (11)

$\text{Re}\ \gamma^2 = \alpha^2 - \beta^2 = -\omega^2\mu\varepsilon$

$-\text{Re}\ \gamma^2 = -\alpha^2 + \beta^2 = \omega^2\mu\varepsilon$

$\beta^2 - \alpha^2 = \omega^2\mu\varepsilon$ —— (12).

(11) $\Rightarrow$

$$|\gamma^2| = \sqrt{(\omega^2\mu\varepsilon)^2 + (\sigma\omega\mu)^2}$$

$$= \sqrt{\omega^4\mu^2\varepsilon^2 + \sigma^2\omega^2\mu^2}$$

$$= \sqrt{\omega^2\mu^2(\sigma^2 + \omega^2\varepsilon^2)}$$

$$|\gamma^2| = \omega\mu\sqrt{\sigma^2 + \omega^2\varepsilon^2} \quad \text{——} (13)$$

$(\beta^2 + \alpha^2)$

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Solve eqn (12) & (13)

$$\beta^2 - \alpha^2 = \omega^2\mu\varepsilon$$

$$\beta^2 + \alpha^2 = \omega\mu\sqrt{\sigma^2 + \omega^2\varepsilon^2}$$

$$2\beta^2 = \omega^2\mu\varepsilon + \omega\mu\sqrt{\sigma^2 + \omega^2\varepsilon^2}$$

$$\beta = \frac{1}{2}\left(\omega^2\mu\varepsilon + \omega\mu\sqrt{\sigma^2 + \omega^2\varepsilon^2}\right)$$

$$\beta = \sqrt{\frac{1}{2}\left(\omega^2\mu\varepsilon + \omega\mu\sqrt{\sigma^2 + \omega^2\varepsilon^2}\right)}$$

$$= \sqrt{\frac{1}{2}\left(\omega^2\mu\varepsilon + \omega\mu \cdot \omega\varepsilon\sqrt{1 + \frac{\sigma^2}{\omega^2\varepsilon^2}}\right)}$$

$$= \sqrt{\frac{1}{2}\left(\omega^2\mu\varepsilon + \omega^2\mu\varepsilon\sqrt{1 + \frac{\sigma^2}{\omega^2\varepsilon^2}}\right)}$$

$$= \omega\sqrt{\frac{\mu\varepsilon}{2}\left\{1 + \sqrt{1 + \frac{\sigma^2}{\omega^2\varepsilon^2}}\right\}}$$

$$\beta = \omega\sqrt{\frac{\mu\varepsilon}{2}\left\{1 + \sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2}\right\}} \quad \text{---(14)}$$

The quantity β is a measure of the phase shift per unit length in radians per meter & is called the phase constant or wave number.

Similarly

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left\{\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right\}} \quad \text{---(15)}$$

$\alpha \rightarrow$ attenuation constant or attenuation coefficient
↳ is a measure of the spatial rate of decay of the wave in the medium,
↳ measured in nepers per meter Np/m.
↳ Expressed in decibels per meter dB/m.

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If we assume that the wave propagates along $+a_z$ and that E_s has only an x-component, then

$$E_s = E_{xs}(z)\,a_x \quad \text{——} (16)$$

Substitute (16) in (6)

$$\nabla^2 E_{xs}(z)\,a_x - \gamma^2 E_{xs}(z)\,a_x = 0$$

$$(\nabla^2 - \gamma^2)\,E_{xs}(z) = 0$$

Hence

$$\frac{\partial^2 E_{xs}(z)}{\partial x^2} + \frac{\partial^2 E_{xs}(z)}{\partial y^2} + \frac{\partial^2 E_{xs}(z)}{\partial z^2} - \gamma^2 E_{xs}(z) = 0$$

($\partial^2/\partial x^2 = 0$, $\partial^2/\partial y^2 = 0$)

$$\frac{\partial^2 E_{xs}(z)}{\partial z^2} - \gamma^2 E_{xs}(z) = 0$$

$$\left[\frac{\partial^2}{\partial z^2} - \gamma^2\right] E_{xs}(z) = 0 \quad \text{— (i)}$$

This is a scalar wave equation ⟶

Intrinsic Impedance is given by the formula

$$|\eta| = \frac{\sqrt{\mu/\epsilon}}{\left[1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2\right]^{1/4}} \quad \text{—(17)}$$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$$

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→ A linear homogeneous differential eqn with solution

$$E_{xs}(z) = E_0 e^{-\gamma z} + E_0' e^{\gamma z} \quad \text{— (ii)}$$

$E_0, E_0' \rightarrow$ constants.

field must be finite at infinity requires that $E_0' = 0$.

$e^{\gamma z} \rightarrow$ A wave traveling along $-az$ whereas we assume wave propagation along az, $E_0' = 0$

Inserting the time factor $e^{j\omega t}$ into equation (ii) & use eqn (9), we obtain

$$E(z,t) = Re\left[E_{xs}(z)\, e^{j\omega t}\, a_x\right]$$

$$= Re\left[E_0 e^{-\alpha z}\, e^{j(\omega t - \beta z)}\, a_x\right]$$

or

$$\boxed{E(z,t) = E_0 e^{-\alpha z} \cos(\omega t - \beta z)\, a_x}$$

$$H(z,t) = Re\left[H_0 e^{-\alpha z}\, e^{j(\omega t - \beta z)}\, a_y\right)$$

where $H_0 = \frac{E_0}{\eta}$

$\eta \rightarrow$ complex quantity known as the intrinsic impedance in ohms.

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = |\eta| \angle\theta_\eta = |\eta| e^{j\theta_\eta}$$

with $\boxed{|\eta| = \frac{\sqrt{\mu/\varepsilon}}{\left[1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2\right]^{1/4}}, \quad \tan 2\theta_\eta = \frac{\sigma}{\omega\varepsilon}}$

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where $0 \leq \theta_\eta \leq 45^\circ$.

$$\therefore H = Re\left[\frac{E_0}{|\eta| e^{j\theta_\eta}} e^{-\alpha z} e^{j(\omega t - \beta z)} a_y\right]$$

$$H = \frac{E_0}{|\eta|} e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta) a_y$$

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$\tan\theta$ → used to determine how lossy a medium is

$$\tan\theta = \frac{\sigma}{\omega\varepsilon} \quad \text{—— (18)}$$

↳ loss tangent

A medium is said to be good/perfect (lossless) dielectric if very $\tan\theta \to$ small & $\sigma \ll \omega\varepsilon$

good conductor very $\tan\theta \to$ large & $\sigma \gg \omega\varepsilon$

$\theta \to$ loss angle of the medium.

The complex permitivity of the medium is given by

$$\nabla \times H_s = (\sigma + j\omega\varepsilon)E_s$$
$$= j\omega\varepsilon\left(1 - \frac{j\sigma}{\omega\varepsilon}\right)E_s$$
$$= j\omega\varepsilon_c E_s$$

where $\varepsilon_c = \varepsilon\left(1 - j\frac{\sigma}{\omega\varepsilon}\right)$

$$\varepsilon_c = \varepsilon\left[1 - j\frac{\sigma}{\omega\varepsilon}\right] \quad \text{—— (19)}$$

or

$$\varepsilon_c = \varepsilon' - j\varepsilon'' \quad \text{—— (20)}$$

where

$$\varepsilon' = \varepsilon$$
$$\varepsilon'' = \frac{\sigma}{\omega}$$

$\varepsilon_c \to$ complex permitivity of the medium.

The ratio of ε'' to ε' is the loss tangent of the medium

$$\tan\theta = \frac{\varepsilon''}{\varepsilon'}$$
$$= \frac{\sigma}{\omega\varepsilon} \quad \text{—— (21)}$$

The wave velocity $u = \frac{\omega}{\beta}$

wavelength $\lambda = \frac{2\pi}{\beta}$

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Plane Waves in Lossless Dielectrics.

In a lossless dielectric,

$$\sigma \ll \omega\varepsilon$$

$$\boxed{\sigma \approx 0 \;,\; \varepsilon = \varepsilon_0\varepsilon_r \;,\; \mu = \mu_0\mu_r} \quad \text{—(22)}$$

Substitute these into equation (14) & (15)

$$\alpha = 0$$

$$\beta = \omega\sqrt{\mu\varepsilon}$$

Vel $u = \dfrac{\omega}{\beta}$

$$= \frac{\omega}{\omega\sqrt{\mu\varepsilon}}$$

Vel $u = \dfrac{1}{\sqrt{\mu\varepsilon}}$

$$\lambda = \frac{2\pi}{\beta}$$

Also w.k.t $\eta = \sqrt{\dfrac{j\omega\mu}{\sigma + j\omega\varepsilon}}$

$$= \sqrt{\frac{j\omega\mu}{j\omega\varepsilon}}$$

$$\eta = \sqrt{\frac{\mu}{\varepsilon}} \angle 0^\circ$$

Thus E and H are in time phase with each other.

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Plane Waves in free Space:

Condition

$$\boxed{\sigma = 0,\ \varepsilon = \varepsilon_0,\ \mu = \mu_0} \quad \text{---(23)}$$

Substitute this in eqn (14) & (15)

$$\alpha = 0$$

$$\beta = \omega\sqrt{\mu_0\varepsilon_0} \qquad C = \frac{1}{\mu_0\varepsilon_0}$$

$$= \frac{\omega}{C}$$

or $u = \frac{\omega}{\beta}$

$$= \frac{\omega}{\omega\sqrt{\mu_0\varepsilon_0}}$$

$$= \frac{1}{\mu_0\varepsilon_0}$$

$V = C$ $\qquad C \approx 3\times10^8$ m/s

speed of light in a vacuum.

$$\lambda = \frac{2\pi}{\beta}$$

EM waves travel in free space at the speed of light.

Substitute eqn (23) in (17)

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi \approx 377\,\Omega.$$

↳ Intrinsic impedance of free space.

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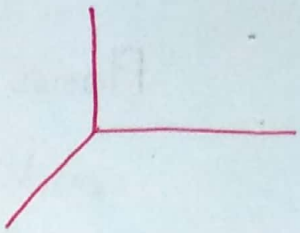
a_E, a_H & a_K are unit vectors along E field, H field & the direction of wave propagation

$a_K \times a_E = a_H$ or $a_K \times a_H = -a_E$ $\boxed{a_E \times a_H = a_K}$

$$E = E_0 \cos(\omega t - \beta z) a_x$$

$$H = H_0 \cos(\omega t - \beta z) a_y$$

$$H = \frac{E_0}{\eta_0} \cos(\omega t - \beta z) a_y$$

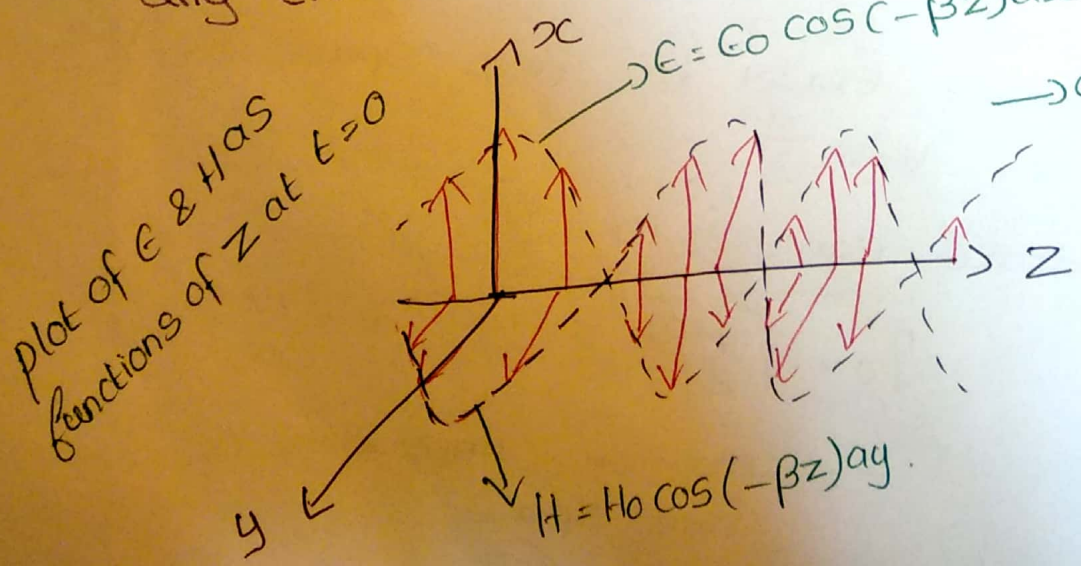


Both E and H fields (or EM waves) are everywhere normal to the direction of wave propagation, a_K.

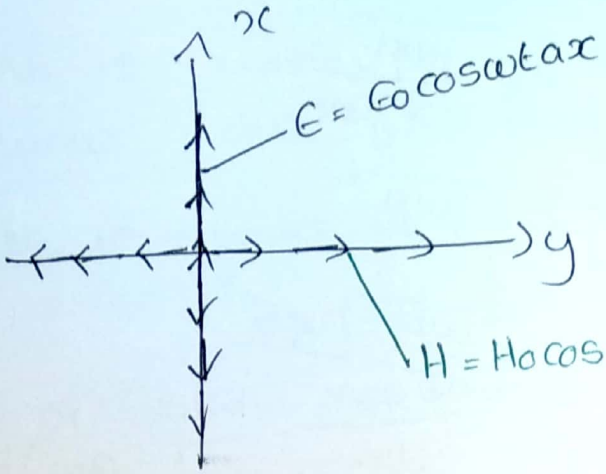
* That means that the fields lie in a plane that is transverse or orthogonal to the direction of wave propagation.

* They form an EM wave that has no electric or magnetic field components along the direction of propagation. Such a wave is called Transverse electromagnetic wave (TEM) wave.

* A combination of E & H is called a uniform plane wave because E (or H) has the same magnitude throughout any transverse plane defined by z = const.



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plot of E and H as functions of z at z=0.

Plane waves in Good Conductors.

A perfect or good conductor is one in which $\sigma >> \omega\varepsilon$, so that $\frac{\sigma}{\omega\varepsilon} >> 1$. (ie)

$$\boxed{\sigma \simeq \infty, \quad \varepsilon = \varepsilon_0, \quad \mu = \mu_0\mu_r} \quad \text{---} (24)$$

Condition

Substitute (24) in (14) & (15)

$$\alpha = \beta = \sqrt{\frac{\omega\mu\sigma}{2}} = \sqrt{\pi f \mu\sigma}$$

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}-1\right]}$$

$\omega = 2\pi f$.

$$u = \frac{\omega}{\beta}$$

$$u = \frac{\omega}{\sqrt{\frac{\omega\mu\sigma}{2}}} = \sqrt{\frac{2\omega}{\mu\sigma}}$$

$$u = \sqrt{\frac{2\omega}{\mu\sigma}}$$

$$\lambda = \frac{2\pi}{\beta}$$

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$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma}}$$

$$\eta = \sqrt{\frac{\omega\mu}{\sigma}} \angle 45^\circ$$

$$E = E_0 e^{-\alpha z} \cos(\omega t - \beta z) a_x$$

$$H = \frac{E_0}{\sqrt{\frac{\omega\mu}{\sigma}}} e^{-\alpha z} \cos(\omega t - \beta z - 45^\circ) a_y .$$

∴ As the E (or H) wave travels in a conducting medium, its amplitude is attenuated by the factor $e^{-\alpha z}$.

Skin depth: or penetration depth.

It is a measure of the depth to which an EM wave can penetrate the medium.

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} = \frac{1}{\alpha}$$

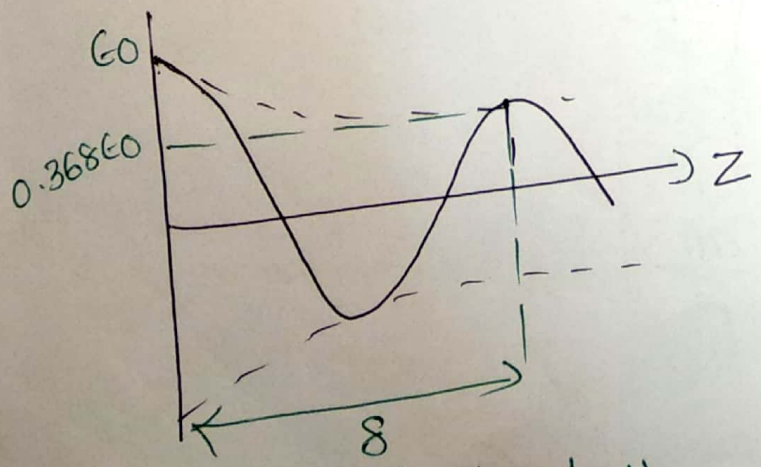


Illustration of skin depth.

The distance δ through which the wave amplitude decreases to a factor e^{-1} (about 37% of the original value)

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Skin depth is useful in calculating the ac resistance due to skin effect.

The dc resistance is $R_{dc} = \frac{l}{\sigma S}$

Surface or skin resistance

$$R_s = \frac{1}{\sigma \delta} = \sqrt{\frac{\pi f \mu}{\sigma}}$$

$$R_{ac} = \frac{l}{\sigma \delta w}$$

$$= \frac{R_s l}{w}$$

for a conductor wire of radius a, $w = 2\pi a$,

$$\frac{R_{ac}}{R_{dc}} = \frac{l/\sigma \delta 2\pi a}{l/\sigma \pi a^2} = \frac{a}{2\delta} = \frac{a}{2}\sqrt{\pi f \mu \delta}$$

$$R_{ac} = \frac{R_s l}{w} = \frac{1}{\sigma \delta} \frac{l}{w} = \frac{l}{\sigma \delta 2\pi a}$$

$$R_{dc} = \frac{l}{\sigma S} = \frac{l}{\sigma \pi a^2}$$

$$\frac{R_{ac}}{R_{dc}} = \frac{l}{\sigma \delta 2\pi a} \times \frac{\sigma \pi a^2}{l}$$

$$= \frac{a}{2\delta}$$

$$= \frac{a}{2\frac{1}{\sqrt{\pi f \mu \delta}}} = \frac{a\sqrt{\pi f \mu \delta}}{2}$$

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Waveguides:

* Waveguides are used at the range of microwave frequencies (3-300GHz) to obtain
  - larger Bandwidth
  &
  - lower signal attenuation.

* Waveguide can operate only above a certain frequency called the cutoff frequency and therefore acts as a high pass filter.

* Thus Waveguides cannot transmit dc and they become excessively large at frequencies below microwave frequencies.

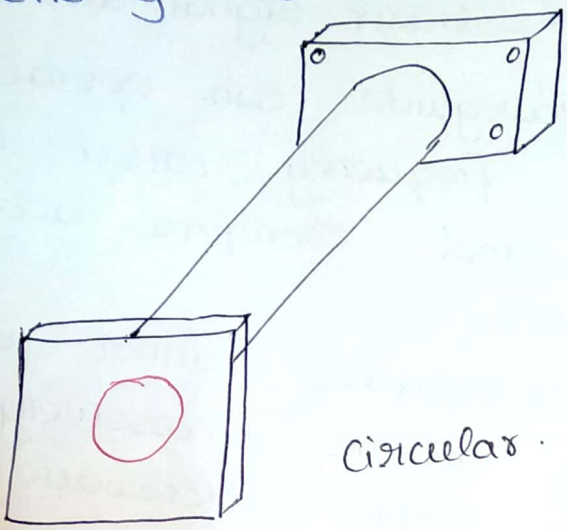
* A waveguide may assume any arbitrary but uniform cross section, common waveguides are either
  - Rectangular
  or
  - circular.

* we will consider only Rectangular waveguides.

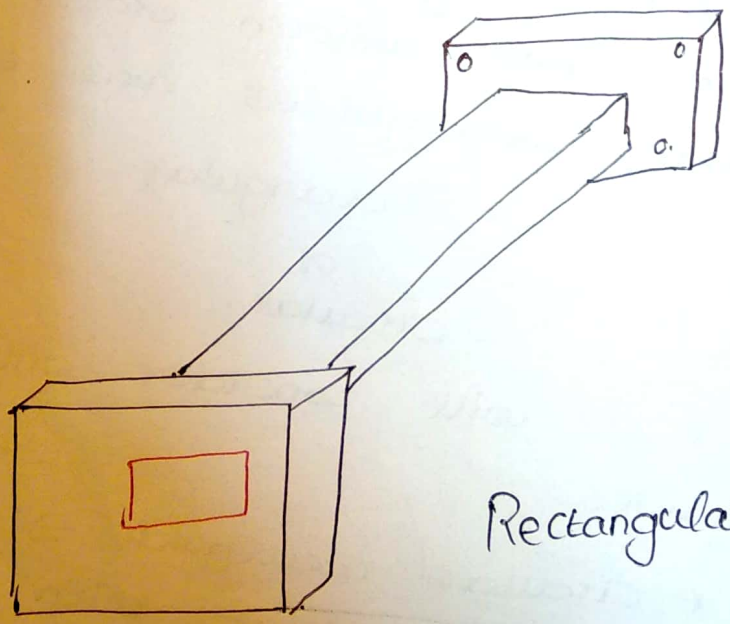
* Circular waveguide → Requires familiarity with Bessel fn. (difficult).

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* By assuming lossless waveguides ($\sigma_c \approx \infty, \sigma \approx 0$) we shall apply maxwell's eqn with appropriate boundary conditions to obtain different modes of wave propagation & the corresponding E and H fields.



Circular.



Rectangular.

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"Praise the Lord"

(10)

(25)

Rectangular WaveGuides

→ Waveguides usually take the form of rectangular or circular cylinders because of its simpler forms and lower cost of manufacture.

→ A hollow conducting metallic tube of uniform cross-secti (rectangular / circular) is used for propagating electromagnetic waves, waves that are guided along the surface (walls) of the tube is called a waveguide.

→ Propagation of waveguide can be considered as a phenomen in which the waves are reflected from wall to wall and hence pass down the waveguide in a zig-zag fashion.

In order to determine the electromagnetic field configuratio within the guide, Maxwell's eqns are solved subject to the appropriate boundary conditions at the walls of the guide. variations in the z direction may be expressed as $e^{-\gamma z}$, $\gamma = \alpha + j\beta$

The Maxwell's eqn for non-conducting medium

$\nabla \times H = \sigma E + j\omega\varepsilon E$

$\because \varepsilon = -j\omega\mu H$ $\quad \because \sigma (\sigma + j\omega\varepsilon) E$

$$\nabla \times H = \begin{vmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = j\omega\varepsilon \left(a_x E_x + a_y E_y + a_z E_z \right)$$

Equating x, y & z components

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = j\omega\varepsilon E_x \quad \text{(i)}$$

$$\frac{\partial H_z}{\partial x} - \frac{\partial H_x}{\partial z} = -j\omega\varepsilon E_y \quad \text{(ii)}$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = j\omega\varepsilon E_z \quad \text{(iii)}$$

} — (1)

(10)

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11) $\nabla \times E = -j\omega\mu H$

$$\nabla \times E = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ E_x & E_y & E_z \end{vmatrix} = -j\omega\mu(\vec{a_x}H_x + \vec{a_y}H_y + \vec{a_z}H_z)$$

Equating x, y & z components.

$$\left.\begin{aligned} \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} &= -j\omega\mu H_x \quad (i) \\ \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} &= -j\omega\mu H_y \quad (ii) \\ \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} &= -j\omega\mu H_z \quad (iii) \end{aligned}\right\} \rightarrow (2)$$

similarly the wave eqns:

$$\left.\begin{aligned} \frac{\partial^2 H}{\partial x^2} + \frac{\partial^2 H}{\partial y^2} + \frac{\partial^2 H}{\partial z^2} &= -\omega^2\mu\varepsilon H \\ \frac{\partial^2 E}{\partial x^2} + \frac{\partial^2 E}{\partial y^2} + \frac{\partial^2 E}{\partial z^2} &= -\omega^2\mu\varepsilon E \end{aligned}\right\} \rightarrow (3)$$

when the variation in the z direction of each of the field componen is

let $H_y = H_y^\circ e^{-\gamma z}$

$$\frac{\partial H_y}{\partial z} = -\gamma H_y^\circ e^{-\gamma z} = -\gamma H_y$$

$$\frac{\partial H_x}{\partial z} = -\gamma H_x$$

$H_x = H_x^\circ e^{-\gamma z}$

$\frac{\partial H_x}{\partial z} = -\gamma H_x^\circ e^{-\gamma z}$

$= -\gamma H_x$

let $E_y = E_y^\circ e^{-\gamma z}$

$$\frac{\partial E_y}{\partial z} = -\gamma E_y \quad \& \quad \frac{\partial E_x}{\partial z} = -\gamma E_x$$

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Substituting these values in eqns ① , ② & ③ (26)

$$\frac{\partial H_z}{\partial y} + \gamma H_y = j\omega\varepsilon E_x \quad \text{(i)}$$

$$\frac{\partial H_z}{\partial x} + \gamma H_x = -j\omega\varepsilon E_y \quad \text{(ii)} \qquad \text{——— (4)}$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = j\omega\varepsilon E_z \quad \text{(iii)}$$

$$\frac{\partial E_z}{\partial y} + \frac{\partial E_y}{\partial z} + \gamma E_y = -j\omega\mu H_x \quad \text{(i)}$$

$$\frac{\partial E_z}{\partial x} + \gamma E_x = j\omega\mu H_y \quad \text{(ii)} \qquad \text{— (5)}$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -j\omega\mu H_z \quad \text{(iii)}$$

Wave eqns thus become

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \gamma^2 E_z = -\omega^2\mu\varepsilon E_z$$

$$\frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + \gamma^2 H_z = -\omega^2\mu\varepsilon H_z$$

Solving the eqns 4 (ii) & 5 (i) E_y

$$\frac{\partial H_z}{\partial x} + \gamma H_x = -j\omega\varepsilon E_y \qquad \text{——— 4 (ii)}$$

$$\frac{\partial E_z}{\partial y} + \frac{\partial E_y}{\partial z} + \gamma E_y = -j\omega\mu H_x \qquad \text{——— 5 (i)}$$

$$H_x = -\frac{1}{j\omega\mu}\left[\frac{\partial E_z}{\partial y} + \gamma E_y\right]$$

sub H_x in eqn 4 (ii)

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$$\frac{\partial H_z}{\partial x} + \gamma H_x = -j\omega\xi E_y$$

$$\frac{\partial H_z}{\partial x} + \gamma\left[-\frac{1}{j\omega\mu}\left(\frac{\partial E_z}{\partial y} + \gamma E_y\right)\right] = -j\omega\xi E_y$$

$$\frac{\partial H_z}{\partial x} - \frac{\gamma}{j\omega\mu}\frac{\partial E_z}{\partial y} - \frac{\gamma^2 E_y}{j\omega\mu} = -j\omega\xi E_y$$

$$\frac{\partial H_z}{\partial x} - \frac{\gamma}{j\omega\mu}\frac{\partial E_z}{\partial y} = \left(\frac{\gamma^2}{j\omega\mu} - j\omega\xi\right) E_y$$

multiply $(j\omega\mu)$

$$j\omega\mu\frac{\partial H_z}{\partial x} - \gamma\frac{\partial E_z}{\partial y} = \gamma^2 - (j\omega\mu)(j\omega\xi) E_y$$

$$= \left(\gamma^2 + \omega^2\mu\xi\right) E_y$$

$$j\omega\mu\frac{\partial H_z}{\partial x} - \gamma\frac{\partial E_z}{\partial y} = h^2 E_y$$

$$\boxed{E_y = \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x} - \frac{\gamma}{h^2}\frac{\partial E_z}{\partial y}}$$

Now H_x

$$H_x = -\frac{1}{j\omega\mu}\left(\frac{\partial E_z}{\partial y} + \gamma E_y\right)$$

$$= -\frac{1}{j\omega\mu}\left(\frac{\partial E_z}{\partial y} + \gamma\left(\frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x} - \frac{\gamma}{h^2}\frac{\partial E_z}{\partial y}\right)\right)$$

$$= \frac{-1}{j\omega\mu}\frac{\partial E_z}{\partial y} - \frac{\gamma}{h^2}\frac{\partial H_z}{\partial x} + \frac{\gamma^2}{j\omega\mu h^2}\frac{\partial E_z}{\partial y}$$

$$= \frac{-\gamma}{h^2}\frac{\partial H_z}{\partial x} + \frac{-h^2\frac{\partial E_z}{\partial y} + \gamma^2\frac{\partial E_z}{\partial y}}{j\omega\mu h^2}.$$

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$$= \frac{-\gamma}{h^2}\frac{\partial H_z}{\partial x} + \frac{-k^2 + (k^2 + \omega^2\mu\xi)}{j\omega\mu h^2}\frac{\partial E_z}{\partial y} \quad (27)$$

$$= " + \frac{-\omega^2\mu\xi}{j\omega\mu h^2}\frac{\partial E_z}{\partial y}$$

$$= " + \frac{(j\omega\xi)(j\omega\mu)}{j\omega\mu h^2}\frac{\partial E_z}{\partial y}$$

$$\boxed{H_x = \frac{-\gamma}{h^2}\frac{\partial H_z}{\partial x} + \frac{j\omega\xi}{h^2}\frac{\partial E_z}{\partial y}}$$

Solving the equations 4(i) & 5(ii) H_y

$$\frac{\partial H_z}{\partial y} + \gamma H_y = j\omega\xi E_x \quad \text{— 4(i)}$$

$$\frac{\partial E_z}{\partial x} + \gamma E_x = j\omega\mu H_y \quad \text{— 5(ii)}$$

$$4(i) \Rightarrow E_x = \frac{1}{j\omega\xi}\left(\frac{\partial H_z}{\partial y} + \gamma H_y\right)$$

sub E_x in eqn 5(ii)

$$\frac{\partial E_z}{\partial x} + \gamma\left(\frac{1}{j\omega\xi}\left(\frac{\partial H_z}{\partial y} + \gamma H_y\right)\right) = j\omega\mu H_y$$

$$\frac{\partial E_z}{\partial x} + \frac{\gamma}{j\omega\xi}\frac{\partial H_z}{\partial y} + \frac{\gamma^2}{j\omega\xi}H_y = j\omega\mu H_y$$

$$\frac{\partial E_z}{\partial x} + \frac{\gamma}{j\omega\xi}\frac{\partial H_z}{\partial y} = \left(j\omega\mu - \frac{\gamma^2}{j\omega\xi}\right)H_y \Rightarrow \left(\frac{-\omega^2\mu\xi}{j\omega\xi}\right.$$

(17)

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$\times^{ly}$ by $j\omega\xi$

$$j\omega\xi \frac{\partial E_z}{\partial x} + \gamma \frac{\partial H_z}{\partial y} = -(\omega^2\mu\xi + \gamma^2) H_y$$

$$= -h^2 H_y$$

$$\boxed{H_y = -\frac{j\omega\xi}{h^2} \frac{\partial E_z}{\partial x} - \frac{\gamma}{h^2} \frac{\partial H_z}{\partial y}}$$

similarly

$$E_x = \frac{1}{j\omega\xi} \left(\frac{\partial H_z}{\partial y} + \gamma H_y \right)$$

$$= \frac{1}{j\omega\xi} \left(\frac{\partial H_z}{\partial y} + \gamma \left(-\frac{j\omega\xi}{h^2} \frac{\partial E_z}{\partial x} - \frac{\gamma}{h^2} \frac{\partial H_z}{\partial y} \right) \right.$$

$$= \frac{1}{j\omega\xi} \frac{\partial H_z}{\partial y} - \frac{\gamma}{h^2} \frac{\partial E_z}{\partial x} - \frac{\gamma^2}{h^2 j\omega\xi} \frac{\partial H_z}{\partial y}$$

$$= \frac{-\gamma}{h^2} \frac{\partial E_z}{\partial x} + \left(\frac{h^2 - \gamma^2}{h^2 j\omega\xi} \right) \frac{\partial H_z}{\partial y}$$

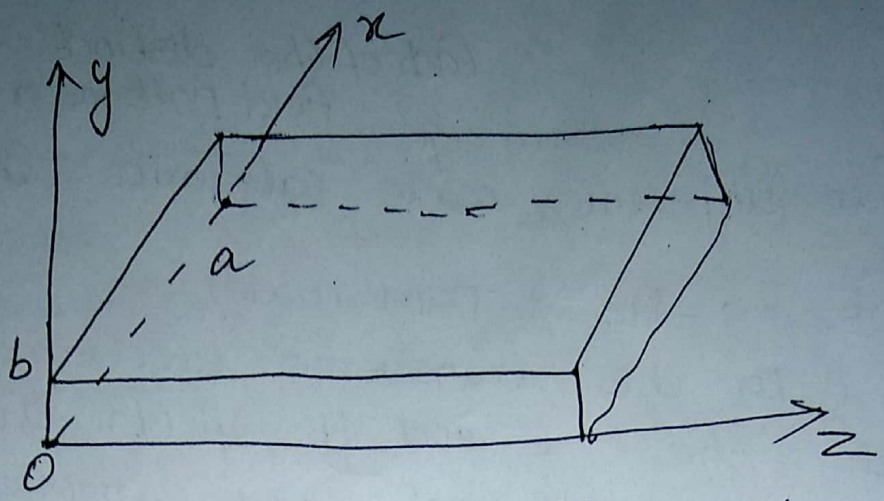
$$= -\gamma/h^2 \frac{\partial E_z}{\partial x} + \frac{h^2 - (h^2 - \omega^2\mu\xi)}{h^2 j\omega\xi} \frac{\partial H_z}{\partial}$$

$$= \eta + + \frac{\omega^2\mu\xi}{h^2 j\omega\xi} \frac{\partial H_z}{\partial y}$$

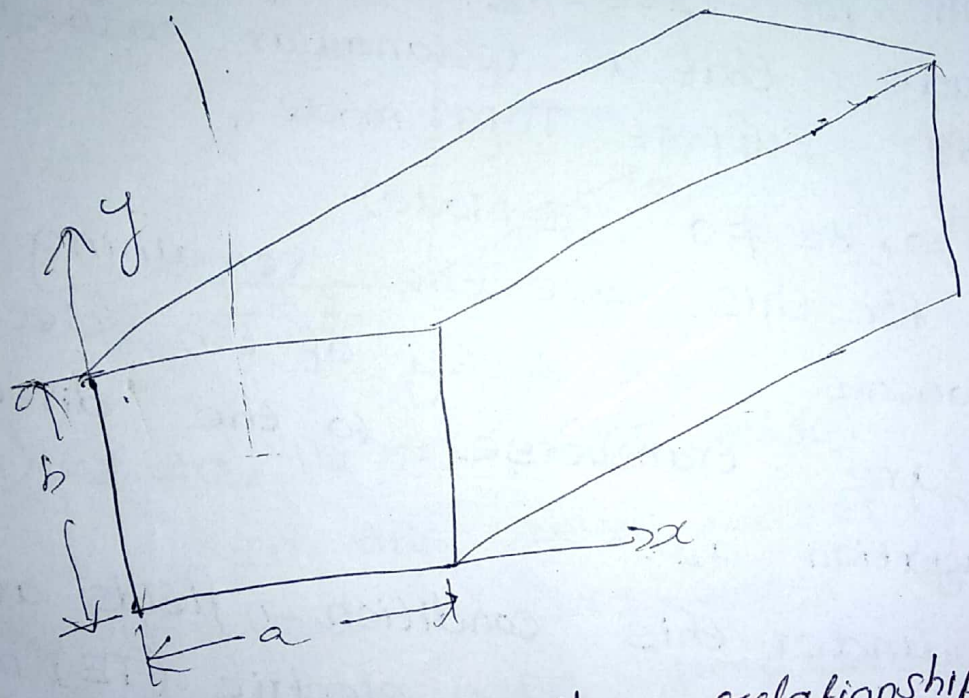
$$= \eta + \frac{(-j\omega\mu)(j\omega\xi)}{h^2 j\omega\xi} \frac{\partial H_z}{\partial y}$$

$$\boxed{E_x = \frac{-\gamma}{1.2} \frac{\partial E_z}{\partial} + -j\omega\mu \frac{\partial H_z}{\partial}}$$

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A rectangular waveguide.



These equations gives the relationships among the fields in the guide.

If E_z & H_z both are zero, all fields within the guide will vanish. Therefore, for waveguide action there must exist either an E_z (or) an H_z component.

Boundary conditions.

$E_x = E_z = 0$ at $y = 0$ & $y = b$

$E_y = E_z = 0$ at $x = 0$ & $x = a$.

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Each of the distinct field patterns or configurations.

Four different mode categories can exist

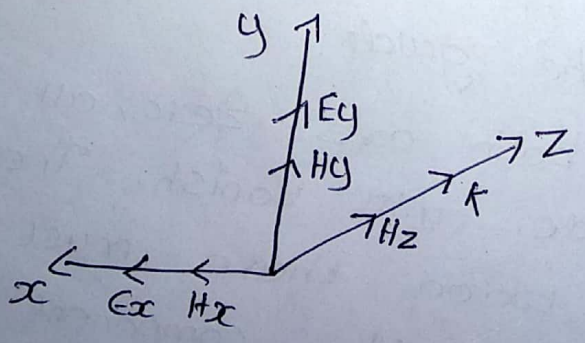
1) $E_z = 0 = H_z$ (TEM mode)

In the transverse electromagnetic mode, both the E and H fields are transverse to the direction of wave propagation. from the eqns we find that all field components vanish for $E_z = 0 = H_z$. consequently we conclude that a rectangular waveguide cannot support TEM mode.

2) $E_z = 0$, $H_z \neq 0$ (TE Mode)

For this case the remaining components E_x & E_y of the electric field are transverse to the direction of propagation az.

Under this condition, fields are said to be in transverse electric (TE) modes.



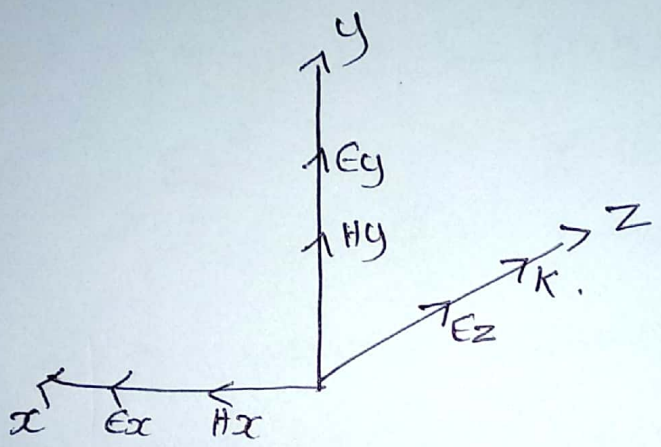
TE mode
$E_z = 0$

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3) $E_z \neq 0, H_z = 0$ (TM modes)

In this case, the H field is transverse to the direction of wave propagation.

Thus we have transverse Magnetic modes. (TM)



TM mode
$H_z = 0$

4) $E_z \neq 0, H_z \neq 0$ (HE Mode).

In this case neither the E nor the H field is transverse to the direction of wave propagation.

Sometimes these modes are referred to as hybrid modes.

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Transverse Magnetic (TM) Waves in Rectangular Guides:

Wave equations Ex, Ey, Hx & Hy of Rectangular waveguide are partial differential equations that can be solved by usual technique of assuming a product solution.

$$E_z(x,y,z) = E_z^0(x,y)\, e^{-\gamma z} \quad \text{— (1)}$$

$$E_z^0 = XY \quad \text{— (2)}$$

X is a function of x
Y is a function of y.

Sub (2) in wave equations,

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \gamma^2 E_z = -\omega^2 \mu\varepsilon E_z$$

$$\frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + \gamma^2 H_z = -\omega^2 \mu\varepsilon H_z$$

} wave eqns.

$$Y\frac{\partial^2 X}{\partial x^2} + X\frac{\partial^2 Y}{\partial y^2} + \gamma^2 XY = -\omega^2 \mu\varepsilon XY$$

$$Y\frac{\partial^2 X}{\partial x^2} + X\frac{\partial^2 Y}{\partial y^2} + XY\left(\gamma^2 + \omega^2\mu\varepsilon\right) = 0$$

$$\div XY \quad \frac{1}{X}\frac{\partial^2 X}{\partial x^2} + \frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} + h^2 = 0$$

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} + h^2 = \frac{-1}{Y}\frac{\partial^2 Y}{\partial y^2} \quad \text{— (3)}$$

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eqn ③ equates a function of x alone to a function of y alone.

Only way to hold the relation, these functions equation to some constant say A^2

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} + h^2 = A^2 \quad \text{—— ④}$$

$$③ \Rightarrow \frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} = -A^2 \quad \text{—— ⑤}.$$

Soln of eqn ④ $\quad \frac{1}{X}\frac{\partial^2 X}{\partial x^2} + B^2 = 0$

$$X = C_1 \cos Bx + C_2 \sin Bx \quad \text{—— ⑥}$$

$$B^2 = h^2 - A^2$$

Soln of eqn ⑤ $\quad \frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} + A^2 = 0$

$$Y = C_3 \cos Ay + C_4 \sin Ay \quad \text{—— ⑦}$$

$$E_z^0 = XY$$

$$= (C_1 \cos Bx + C_2 \sin Bx)(C_3 \cos Ay + C_4 \sin Ay)$$

$$= C_1C_3 \cos Bx \cos Ay + C_1C_4 \cos Bx \sin Ay + C_2C_3 \sin Bx \cos Ay + C_2C_4 \sin Bx \sin Ay. \quad \text{—— ⑧}$$

Constants C_1, C_2, C_3, C_4, A and B must now be selected to fit the boundary conditions.

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$E_z^0 = 0$ when $x = 0, x = a$

$y = 0, y = b$

when $x = 0$ eqn (8) becomes

$$E_z^0 = C_1 C_3 \cos Ay + C_1 C_4 \sin Ay \quad \text{—— (9)}$$

for E_z^0 to vanish (for all values of y)

it is evident that C_1 must be zero.

Sub $C_1 = 0$ in (8)

$$\therefore E_z^0 = C_2 C_3 \sin Bx \cos Ay + C_2 C_4 \sin Bx \sin Ay. \quad \text{— (9)}$$

when $y = 0$ eqn (9) becomes

$$E_z^0 = C_2 C_3 \sin Bx$$

For this to be zero for all values of x, it is possible to have either C_2 or C_3 equal to zero (assuming $B \neq 0$)

C_2 or $C_3 = 0$.

put $C_2 = 0$ in (9)

$E_z^0 = 0$

put $C_3 = 0$ in (9)

$E_z^0 \neq 0$.

$\therefore$ Sub $C_2 = 0$ in eqn (9)

$$E_z^0 = C_2 C_4 \sin Bx \sin Ay \quad \text{—— (10)}$$

$C_2 C_4 = C$ → amplitude Constant.

$$E_z^0 = C \sin Bx \sin Ay. \quad \text{—— (11)}$$

(10)

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To find unknown constants A & B :-

Boundary condition

when $x = a$, $E_z^0 = 0$

Sub this in eqn (11)

$$E_z^0 = C \sin Ba \sin Ay$$

$$\sin Ba = 0$$

$$Ba = \sin^{-1}(0)$$

$$B = \frac{m\pi}{a} \quad , m = 1, 2, 3 \ldots \quad \text{— (12)}$$

when $y = b$, $E_z^0 = 0$

Sub this in eqn (11)

$$E_z^0 = C \sin \frac{m\pi x}{a} \sin Ab$$

$$\sin Ab = 0$$

$$Ab = \sin^{-1}(0)$$

$$A = \frac{n\pi}{b} \quad , n = 1, 2, 3 \ldots \quad \text{— (13)}$$

$\therefore$ (11) $\Rightarrow$

$$E_z^0 = C \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \quad \text{— (14)}$$

Making use of eqns E_x, E_y, H_x, H_y of Rectangular waveguide & putting $\gamma = j\beta$ for frequencies above f_c, the following equations are obtained,

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$$E_x^0 = \frac{-j\beta}{h^2} CB \cos Bx \sin Ay$$

$$E_y^0 = \frac{-j\beta}{h^2} CA \sin Bx \cos By$$

$$H_x^0 = \frac{j\omega\varepsilon}{h^2} CA \sin Bx \cos Ay$$

$$H_y^0 = \frac{-j\omega\varepsilon}{h^2} CB \cos Bx \sin Ay \quad \text{——— (15)}$$

where $A = \frac{n\pi}{b}$, $B = \frac{m\pi}{a}$

Each set of integers m and n gives a different field pattern or mode referred to as TM_{mn} mode, in the waveguide.

Integer m → equals the number of half cycle variations in the x-direction.

Integer n → equals the number of half cycle variations in the y direction.

from eqn (14) & (15)

If (m,n) is (0,0)
(0,n)
or
(m,0)
all field components vanish.

(17)

(18)

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Thus Neither m nor n can be zero.

TM_{11} is the lowest order mode of all the TM_{mn} modes.

To find γ:-

we know that $\gamma^2 + \omega^2\mu\varepsilon = h^2$, $B^2 = h^2 - A^2$

$h^2 = A^2 + B^2$

$$\gamma^2 = h^2 - \omega^2\mu\varepsilon$$

$$\gamma = \sqrt{h^2 - \omega^2\mu\varepsilon}$$

$$= \sqrt{A^2 + B^2 - \omega^2\mu\varepsilon}$$

$$\gamma = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2\mu\varepsilon} \quad \text{---(16)}$$

Case (1) (cutoff)

If $\omega^2\mu\varepsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$

$\gamma = 0$ or $\alpha = 0 = \beta$

The value of ω that causes this is called cutoff angular frequency ω_c,

(ie) $\omega_c^2 = \frac{1}{\mu\varepsilon}\left\{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right\}$

$$\boxed{\omega_c = \frac{1}{\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}} \quad \text{---(17)}$$

No propagation takes place at this frequency

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Case (2) (Evanescent).

$$\text{If } \omega^2\mu\varepsilon < \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$\gamma = \alpha, \quad \beta = 0$$

In this case, we have no wave propagation at all. These nonpropagating modes are said to be evanescent

Case (3) (Propagation)

$$\text{If } \omega^2\mu\varepsilon > \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$\gamma = j\beta, \quad \alpha = 0$$

(16) becomes

$$\therefore \beta = \sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2} \quad \text{---(18)}$$

This is the only case in which propagation takes place because all field components will have the factor

$$e^{-\gamma z} = e^{-j\beta z}.$$

Thus for each mode, characterized by a set of integers m & n, there is a corresponding cutoff frequency fc.

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Cutoff Frequency:

It is the operating frequency below which attenuation occurs and above which propagation takes place.

The waveguide therefore operates as a high-pass filter.

$\omega_c = 2\pi f_c$.

Cutoff frequency $f_c = \frac{\omega_c}{2\pi}$

Sub eqn (17)

$$f_c = \frac{1}{2\pi\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

$$f_c = \frac{u'}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

where $u' = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{\text{phase velocity}}{}$

$4\pi \times 10^{-9} \times 8.854 \times 10^{-12}$ $u' = c = 3 \times 10^8$ m/s

Cutoff wavelength $\lambda_c = \frac{u'}{f_c}$

or

$$\lambda_c = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

TM11 has the lowest cutoff frequency (or the longest cutoff wavelength) of all the TM modes.

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phase constant β in terms of f_c can be written as

eqn (18)

$$\beta = \omega\sqrt{\mu\varepsilon}\sqrt{1-\left(\frac{f_c}{f}\right)^2}$$

or

$$\boxed{\beta = \beta'\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

where $\beta' = \frac{\omega}{u'}$

$\beta' = \omega\cdot\sqrt{\mu\varepsilon}$ = phase constant of uniform plane wave in the dielectric medium.

γ for Evanescent mode in terms of f_c:-

$$\boxed{\gamma = \alpha = \beta'\sqrt{\left(\frac{f_c}{f}\right)^2-1}}$$

phase velocity:-

$$\boxed{u_p = \frac{\omega}{\beta}}$$

wavelength:-

$$\boxed{\lambda = \frac{2\pi}{\beta} = \frac{u_p}{f}}$$

$\lambda = \frac{2\pi}{\omega} \times u_p$

$= \frac{2\pi \times u_p}{2\pi f}$

$\lambda = \frac{u_p}{f}$

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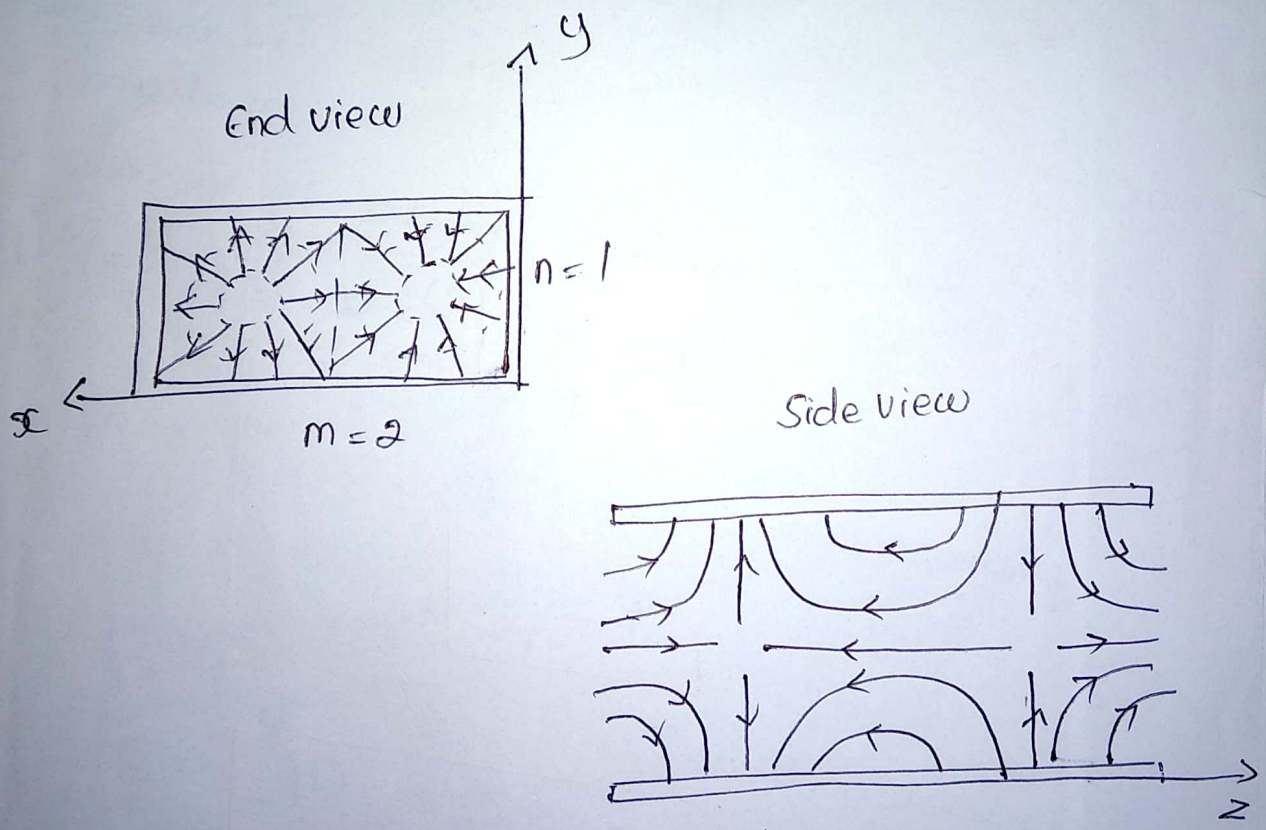
Intrinsic impedance :-

Obtained from eqn (15) as $\gamma = j\beta$.

$$\eta_{TM} = \sqrt{\frac{\mu}{\varepsilon}} \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

$$\eta_{TM} = \eta' \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

where $\eta' = \sqrt{\mu/\varepsilon}$



Field configuration for TM_{21} mode.

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Transverse electric waves in Rectangular waveguide.

In the TE mode, electric field is transverse (or normal) to the direction of wave propagation.

$$E_z = 0 \quad , H_z \neq 0$$

$$H_z^0 = C \cos Bx \cos Ay$$

$$H_x^0 = \frac{j\beta}{h^2} CB \sin Bx \cos Ay$$

$$H_y^0 = \frac{j\beta}{h^2} CA \cos Bx \sin Ay$$

$$E_x^0 = \frac{j\omega\mu}{h^2} CA \cos Bx \sin Ay$$

$$E_y^0 = \frac{-j\omega\mu}{h^2} CB \sin Bx \cos Ay .$$

where $B = \frac{m\pi}{a}$

$$A = \frac{n\pi}{b}.$$

$$E_x = \frac{j\omega\mu}{h^2} C \left(\frac{n\pi}{b}\right) \cos \frac{m\pi x}{a} \cdot \sin \frac{n\pi y}{b} \cdot e^{-\gamma z}$$

$$E_y = -\frac{j\omega\mu}{h^2} C \left(\frac{m\pi}{a}\right) \sin \frac{m\pi x}{a} \cdot \cos \frac{n\pi y}{b} \cdot e^{-\gamma z}$$

$$H_y = \frac{j\beta}{h^2} C \left(\frac{n\pi}{b}\right) \cos \frac{m\pi x}{a} \cdot \sin \frac{n\pi y}{b} \cdot e^{-\gamma z}$$

$$H_x = \frac{j\beta}{h^2} C \left(\frac{m\pi}{a}\right) \sin \left(\frac{m\pi}{a}\right) x \cdot \cos \left(\frac{n\pi}{b}\right) y \cdot e^{-\gamma z}$$

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$E_z = E_z^0 e^{-\gamma z}$, $E_x = E_x^0 e^{-\gamma z}$, $E_y = E_y^0 e^{-\gamma z}$ 6

$H_x = H_x^0 e^{-\gamma z}$, $H_y = H_y^0 e^{-\gamma z}$.

∴ eqn (15) can be written as

$$E_x = \frac{-j\beta}{h^2} C \cdot \left(\frac{m\pi}{a}\right) \underbrace{E_x^0}_{C} \cos Bx \sin Ay \cdot e^{-\gamma z}$$

$$E_y = \frac{-j\beta}{h^2} C \cdot \left(\frac{n\pi}{b}\right) \sin Bx \cos By \cdot e^{-\gamma z}$$

$$H_x = \frac{j\omega\varepsilon}{h^2} C \cdot \left(\frac{n\pi}{b}\right) \sin Bx \cos Ay \cdot e^{-\gamma z}$$

$$H_y = \frac{-j\omega\varepsilon}{h^2} C \cdot \left(\frac{m\pi}{a}\right) \cos Bx \sin Ay \cdot e^{-\gamma z}$$

where $A = \frac{n\pi}{b}$

$$B = \frac{m\pi}{a}.$$

$$E_z = C \sin\left(\frac{m\pi}{a}\right)x \cdot \sin\left(\frac{n\pi}{b}\right)y \cdot e^{-\gamma z}.$$

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$m = 0, 1, 2 \cdots$

$n = 0, 1, 2 \cdots$

Cutoff frequency f_c,
Cutoff wavelength λ_c
Phase constant β
Phase velocity u_p
Wavelength λ

} Same as that of TM mode.

For TE modes

(m, n) may be $(0,1)$ or $(1,0)$

but not

$(0,0)$

Both m & n cannot be zero at the same time because

↳ it will force the field components to vanish.
(E_x, E_y, H_x, H_y)

∴ The lowest mode can be TE_{10} or TE_{01} depending on the values of a & b, the dimension of the guide

It is standard practice to have

$a > b$

So that $\frac{1}{a^2} < \frac{1}{b^2}$ in eqn $f_c = \frac{u'}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$

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Thus TE_{10} is the lowest mode.

because $f_{c_{TE10}} = \frac{u'}{2a} < f_{c_{TE01}} = \frac{u'}{2b}$

This mode is called as dominant mode of the waveguide.

$$f_{c_{10}} = \frac{u'}{2a}.$$

$$\lambda_{c_{10}} = 2a$$

Dominant mode is the mode with the lowest cutoff frequency or (longest cutoff wavelength)

The intrinsic impedance for the TE mode is

$$\eta_{TE} = \sqrt{\frac{\mu}{\varepsilon}} \cdot \frac{1}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

or

$$\eta_{TE} = \frac{\eta'}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

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Wave Propagation in the Guide:-

The field components of TE & TM waves involve the terms sine or cosine of $\left(\frac{m\pi}{a}\right)x$ or $\left(\frac{n\pi}{b}\right)y$ times $e^{-\gamma z}$.

Since

$$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

a wave within the [redacted] waveguide can be resolved into a combination of plane waves reflected from the waveguide walls.

For eg:

For TE_{10} mode

$$E_y = -\frac{j\omega\mu a}{\pi}\sin\left(\frac{\pi x}{a}\right)e^{-j\beta z}$$

$$= -\frac{\omega\mu a}{2\pi}\left(e^{\frac{j\pi x}{a}} - e^{\frac{-j\pi x}{a}}\right)e^{-j\beta z}$$

$$E_y = \frac{\omega\mu a}{2\pi}\left[e^{-j\beta(z+\pi x/\beta a)} - e^{-j\beta(z-\pi x/\beta a)}\right] \quad \text{— (I)}$$

where C = 1.

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The first term of eqn (I) represents a wave traveling in the positive z direction at an angle

$$\theta = \tan^{-1}\left(\frac{\pi}{\beta a}\right)$$

with the z-axis.

The second term of eqn (I) represents a wave traveling in the +ive z direction at an angle $-\theta$.

The wave component in the z direction has a different wavelength from that of the plane waves. This wavelength along the axis of the guide is called the waveguide wavelength. it is given by

$$\lambda = \frac{\lambda'}{\sqrt{1-\left(\frac{fc}{f}\right)^2}}$$

where $\lambda' = \frac{u'}{f}$

$u' \rightarrow$ phase velocity $u' = \frac{1}{\sqrt{\mu\varepsilon}}$

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As a consequence of the zigzag paths, we have 3 types of velocity

1) medium velocity u'
2) phase velocity u_p
3) Group velocity u_g.

Medium velocity:

$$u' = \frac{1}{\sqrt{\mu\varepsilon}}$$

The phase velocity:

↳ It is the velocity at which loci of constant phase are propagated down the guide.

↳ given by

$$u_p = \frac{\omega}{\beta}.$$

$$u_p = \frac{u'}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

$$u_p \geq u'$$

If $u' = c$, then $u_p >$ speed of light in vacuum.

Group velocity:-

↳ It is the velocity with which the resultant repeated reflected waves are traveling down the guide.

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— It is given by

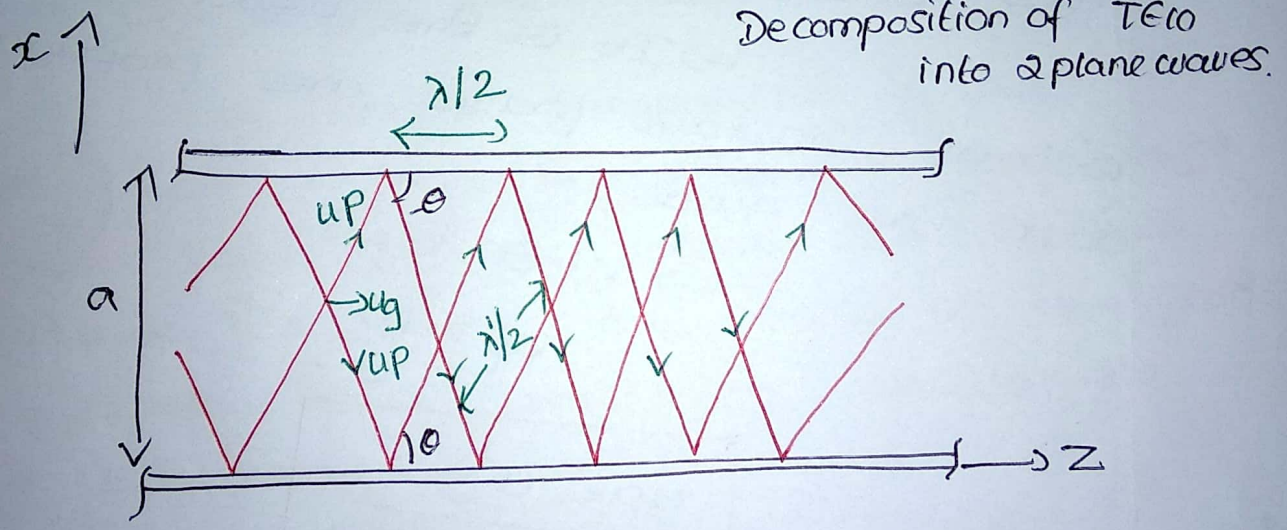
$$u_g = \frac{1}{\partial\beta/\partial\omega}$$

or

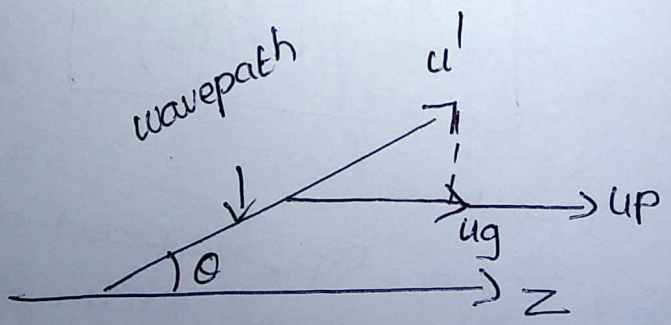
$$u_g = u' \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

Thus:

$$u_p u_g = u'^2$$



Decomposition of TE10 into 2 plane waves.



Relationship between u', up & ug.

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Power Transmission and Attenuation:

The attenuation constants due to dielectric loss is

$$\alpha_d = \frac{\sigma \eta'}{2\sqrt{1-\left(\frac{fc}{f}\right)^2}}$$

$$\eta' = \sqrt{\mu/\varepsilon}$$

The attenuation constant due to ohmic or conduction loss is

$$\alpha_c = \frac{2R_s}{b\eta'\sqrt{1-\left(\frac{fc}{f}\right)^2}}\left(0.5 + \frac{b}{a}\left(\frac{fc}{f}\right)^2\right)$$

$R_s \rightarrow$ Real part of the intrinsic impedance η_c of the conducting wall.

$$\alpha_c\Big|_{TE} = \frac{2R_s}{b\eta'\sqrt{1-\left(\frac{fc}{f}\right)^2}}\left[\left(1+\frac{b}{a}\right)\left(\frac{fc}{f}\right)^2 + \frac{\frac{b}{a}\left(\frac{b}{a}m^2+n^2\right)}{\frac{b^2}{a^2}m^2+n^2}\left(1-\left(\frac{fc}{f}\right)^2\right)\right]$$

$$\alpha_c\Big|_{TM} = \frac{2R_s}{b\eta'\sqrt{1-\left(\frac{fc}{f}\right)^2}}\,\frac{(b/a)^3 m^2+n^2}{(b/a)^2 m^2+n^2}$$

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Power loss per unit length is

$$P_L = R_s H_0^2 \left[b + \frac{a}{2} \left(1 + \frac{\beta^2 a^2}{\pi^2} \right) \right]$$

Total attenuation

$$\alpha = \alpha_c + \alpha_d.$$

Average Power density $P_{av} = \frac{E}{2\eta}$

$$\eta = \sqrt{\frac{\mu}{\varepsilon}}$$

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Problems - unit 3

1) A lossy dielectric has an intrinsic impedance of $200\angle 30^\circ\ \Omega$ at a particular radian frequency ω. If at that frequency, the plane wave propagating through the dielectric has the magnetic field component

$$H = 10\, e^{-\alpha x} \cos\left(\omega t - \frac{1}{2}x\right) a_y \ A/m.$$

find ϵ and α. Determine the skin depth and wave polarization.

$$E = Re\left[E_0\, e^{-\alpha z}\, e^{j(\omega t - \beta z)}\right] a_x.$$

$$H_0 = \frac{E_0}{\eta}$$

$$\Rightarrow E_0 = \eta \cdot H_0$$

$$= 200 e^{\frac{j\pi}{6}} \times 10$$

$$E_0 = 2000\, e^{j\pi/6}$$

$$H_0 = 10$$

$$\eta = 200\angle 30^\circ$$

$$= 200\, e^{j\theta\eta}$$

$$\eta = 200\, e^{\frac{j\pi}{6}}$$

$$E = Re\left[2000\, e^{j\pi/6} \cdot e^{-\alpha z}\, e^{j(\omega t - \beta z)}\right] a_x.$$

Except for the amplitude & phase difference E and H always have the same form.

$$\therefore E = Re\left[2000\, e^{j\pi/6}\, e^{-\alpha x}\, e^{j(\omega t - \beta x)}\right] a_z.$$

$H = 10\, e^{-\alpha x} \cos(\omega t - \frac{1}{2}x) a_y$

$\frac{1}{2} = \beta$.

$$E = Re\left[2000\, e^{-\alpha x}\, e^{j(\omega t + \frac{\pi}{6} - \frac{1}{2}x)}\right] a_z.$$

$a_K = a_x$
$a_H = a_y$
$-a_E = a_K \times a_H = a_x \times a_y = a_z$
$a_E = -a_z$

or

$$E = -2e^{-\alpha x} \cos\left(\omega t - \frac{x}{2} + \frac{\pi}{6}\right) a_z \ KV/m.$$

(1)

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Now we know that $\beta = \frac{1}{2}$

$$\alpha = \omega \sqrt{\frac{\mu\varepsilon}{2}\left\{\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right\}}$$

$$\beta = \omega \sqrt{\frac{\mu\varepsilon}{2}\left\{\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1\right\}}$$

$$\frac{\alpha}{\beta} = \left\{\frac{\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1}{\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1}\right\}^{1/2}$$

$$\tan 2\theta_\eta = \frac{\sigma}{\omega\varepsilon}$$

$$\tan 60 = \frac{\sigma}{\omega\varepsilon}$$

$$\Rightarrow \frac{\sigma}{\omega\varepsilon} = \sqrt{3}.$$

$$\therefore \frac{\alpha}{\beta} = \left[\frac{\sqrt{1+3} - 1}{\sqrt{1+3} + 1}\right]^{1/2} = \left(\frac{2-1}{2+1}\right)^{1/2} = \left(\frac{1}{3}\right)^{1/2}$$

$$= 1/\sqrt{3}.$$

$$\frac{\alpha}{\beta} = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\beta}{\sqrt{3}}$$

$$\alpha = \frac{1}{2\sqrt{3}} = 0.2887 \text{ Np/m}.$$

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Skin depth

$$\delta = \frac{1}{\alpha}$$

$$= \frac{2\sqrt{3}}{2}$$

$$= 3.464 \text{ m}.$$

The wave has only an E_z component, hence it is polarized along the z-direction.

(2) A uniform plane wave propagating in a medium has

$$E = 2e^{-dz}\sin(10^8 t - \beta z)a_y \text{ V/m}.$$

If the medium is characterised by $\varepsilon_r = 1$, $\mu_r = 20$, and $\sigma = 3$ S/m. Find α, β & H.

Soln.

Determine whether the medium is a lossy dielectric or a good conductor.

$$\frac{\sigma}{\omega\varepsilon} = \frac{3}{10^8 \times 8.854 \times 10^{-12}} = 3388 >> 1$$

Showing medium may be regarded as a good conductor.

$$\alpha = \beta = \sqrt{\frac{\omega\mu\sigma}{2}} = \sqrt{\frac{10^8 \times 4\pi \times 10^{-7} \times 20 \times 3}{2}}$$

$$= 61.38$$

$$\alpha = 61.38 \text{ Np/m}$$

$$\beta = 61.38 \text{ rad/m}.$$

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$$H = H_0 e^{-\alpha z} \sin\left(\omega t - \beta z - \frac{\pi}{4}\right) a_H .$$

$\alpha = 61.38$ Np/m

$\omega = 10^8$

$\beta = 61.38$ rad/m.

$$H_0 = \frac{\epsilon_0}{|\eta|}$$

$$|\eta| = \sqrt{\frac{\mu\omega}{\sigma}}$$

$$= \sqrt{\frac{4\pi \times 10^{-7} \times 20 \times 10^8}{3}}$$

$$|\eta| = 28.93$$

$\epsilon_0 = 2$

$$H_0 = \frac{2}{28.93} = 0.0691$$

$a_K \times a_H = -a_E$

$a_H = a_K \times a_E = a_z \times a_y = -a_x$

$$\therefore H = -0.0691 e^{-61.38z} \sin\left(10^8 t - 61.38 z - \frac{\pi}{4}\right) a_x \text{ A/m}$$

or

$$H = -69.1 e^{-61.38z} \sin\left(10^8 t - 61.38 z - \frac{\pi}{4}\right) a_x \text{ mA/m}.$$

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3) In a Rectangular waveguide for which $a = 1.5cm$, $b = 0.8cm$, $\sigma = 0$, $\mu = \mu_0$, $\varepsilon = 4\varepsilon_0$, $\beta = 1$. The waveguide is operating in TM_{13} mode.

$$H_{xc} = 2\sin\left(\frac{\pi x}{a}\right)\cos\left(\frac{3\pi y}{b}\right)\sin(\pi.10^{11}t - \beta z)\ A/m.$$

Determine (i) cutoff frequency (ii) phase constant (iii) Propagation constant.

Soln

TM_{13} mode

$m = 1, n = 3$

$$f_c = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

(i) Cutoff frequency

$$f_c = \frac{u'}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ or}$$

$$u' = \frac{1}{\sqrt{\mu\varepsilon}} = 149.93\times10^6 \quad \hookrightarrow \varepsilon = 4\varepsilon_0$$

$$f_c = \frac{149.93\times10^6}{2}\sqrt{\left(\frac{1}{1.5\times10^{-2}}\right)^2 + \left(\frac{3}{0.8\times10^{-2}}\right)^2}$$

$$= 28.55\ GHz$$

(ii) phase constant

$$\beta = \omega\sqrt{\mu\varepsilon}\sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

$$= \pi.10^{11}.\sqrt{\mu\varepsilon}\sqrt{1 - \left(\frac{28.55\times10^9}{5\times10^{10}}\right)^2}$$

$$\beta = 1719.13\ rad/m$$

$\omega = 2\pi f = \pi.10^{11}$

$f = \frac{\pi.10^{11}}{2\pi}$

$f = 5\times10^{10}$

$= 50\ GHz$

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(iii) Propagation Constant

$$\gamma = j\beta$$

$$\gamma = j1719.13/m.$$

4) A Standard air filled rectangular waveguide with dimensions $a = 8.636cm$ $b = 4.318cm$ is fed by a 4GHz carrier from a coaxial cable. Determine whether a TE10 mode will be propagated. If So calculate the phase velocity & group velocity

Given data: $a = 8.636cm$

$b = 4.318cm$

$f = 4GHz$

TE10 mode.

TE10 → dominant mode

$$f_c = \frac{u'}{2a} = \frac{3\times10^8}{2\times8.636\times10^{-2}} = 1.7369GHz.$$

As $f = 4GHz > f_c$, TE10 will propagate.

phase velocity $u_p = \frac{u'}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}} = \frac{3\times10^8}{\sqrt{1-\frac{1.736\times10^9}{4\times10^9}}}$

$$u_p = 398.8\times10^6 m/s$$

Group velocity $u_g = u'\sqrt{1-\left(\frac{f_c}{f}\right)^2}$ or $u_p u_g = u'^2$

$$u_g = 3\times10^8\sqrt{1-\left(\frac{1.7369}{4}\right)^2} \qquad u_g = \frac{u'^2}{u_p}.$$

$$u_g = 243.49\times10^6 m/s$$

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5) A rectangular waveguide has internal dimensions 2cm x 1cm and is propagating a 10GHz signal in the dominant mode (TE10) Find the guided wavelength and wave impedance.

Dimension = 2cm x 1cm

a = 2cm f = 10GHz

b = 1cm

TE10 mode

m = 1 , n = 0

Guided wavelength $\lambda = \frac{\lambda'}{\sqrt{1-\left(\frac{fc}{f}\right)^2}}$

$\lambda' = \frac{u'}{f}$

$u' = \frac{1}{\sqrt{\mu\varepsilon}} = 3\times10^8$ m/s.

$\lambda' = \frac{3\times10^8}{10\times10^9} = 0.03$

$\lambda = \frac{0.03}{\sqrt{1-\left(\frac{7.5G}{10G}\right)^2}}$

$fc = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2}$.

$= \frac{3\times10^8}{2}\sqrt{\left(\frac{1}{2\times10^{-2}}\right)^2}$

$fc = 7.5GHz$.

$\lambda = 0.0453$ m.

or

$\lambda = 4.53$ cm.

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Wave impedance $\eta_{TE} = \dfrac{\sqrt{\mu/\varepsilon}}{\sqrt{1-\left(\frac{fc}{f}\right)^2}}$ or $\dfrac{\eta'}{\sqrt{1-\left(\frac{fc}{f}\right)^2}}$

$$\eta_{TE} = \frac{\sqrt{4\pi\times10^{-7}/8.854\times10^{-12}}}{\sqrt{1-\left(\frac{7.5G}{10G}\right)^2}}$$

$$\eta_{TE} = 569.423\ \Omega.$$

6) A Rectangular metal wire, TE guide filled with a dielectric material of relative permitivity $\varepsilon_r = 4$ has the inside dimensions 3cm x 1.2cm. find the cutoff frequency for the dominant mode.

TE_{10} mode → dominant mode

$$fc = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2}.$$

$\varepsilon_r = 4$, $a = 3cm$, $b = 1.2cm$, $m = 1$, $n = 0$

$$fc = \frac{1}{2\sqrt{4\pi\times10^{-7}\times8.854\times10^{-12}\times4}}\sqrt{\left(\frac{1}{3\times10^{-2}}\right)^2+0}$$

$$fc = 2.4989\ GHz.$$

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7) A rectangular waveguide of internal dimensions $a = 4\,cm$, $b = 3\,cm$ is to be operated in the TE11 mode. Find the minimum operating frequency.

Soln

TE11 mode

$m = 1, n = 1$

$\mu_0 = 4\pi \times 10^{-7}$

$\varepsilon_0 = 8.854 \times 10^{-12}$.

$$f_c = \frac{1}{2\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$= \frac{1}{2\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{1}{4\times10^{-2}}\right)^2 + \left(\frac{1}{3\times10^{-2}}\right)^2}$$

625 + 1111.11 41.66

$$f_c = 6.246\,GHz.$$

6) An air filled rectangular waveguide has inner dimensions of 3cm x 2cm. Find the wave impedance of the TE20 mode of propagation in the waveguide at a frequency of 30GHz. ($\eta_0 = 377\,\Omega$) → free space impedance.

Soln.

$f = 30GHz$, $m = 2, n = 0$

$a = 3cm, b = 2cm$.

Given:

$\sqrt{\mu/\varepsilon} = 377\,\Omega$.

$$\eta_{TE} = \frac{\sqrt{\mu/\varepsilon}}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

$$\eta_{TE} = \frac{377}{\sqrt{1 - \left(\frac{1\times10^{10}}{30\times10^9}\right)^2}}$$

$\eta_{TE} = 399.8\,\Omega$ or $400\,\Omega$.

$$f_c = \frac{1}{2\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$= \frac{3\times10^8}{2} \sqrt{\left(\frac{2}{3\times10^{-2}}\right)^2 + 0}$$

$f_c = 1\times10^{10}\,Hz$.

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9) A coaxial cable with an inner diameter of 1mm and outer diameter of 2.4mm is filled with a dielectric of relative permitivity 10.89. find the characteristic impedance.

Inner diameter $a = 1mm$

outer diameter $b = 2.4mm$.

$\varepsilon_r = 10.89$

$\varepsilon_0 = 8.854 \times 10^{-12}$ F/m.

$\mu_0 = 4\pi \times 10^{-7}$ H/m

$$Z_0 = \frac{1}{2\pi}\sqrt{\frac{\mu}{\varepsilon}}\ln\left(\frac{b}{a}\right)$$

$$= \frac{1}{2\pi}\sqrt{\frac{4\pi \times 10^{-7}}{8.854 \times 10^{-12} \times 10.89}}\ln\left(\frac{2.4mm}{1mm}\right)$$

0.875

$Z_0 = 15.3\Omega$

10) The electric field component of a time harmonic plane EM wave travelling in a non-magnetic lossless dielectric medium has an amplitude of 1V/m. If the relative permitivity of the medium is 4. find the magnitude of the time average power density vector in w/m^2.

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Given data:-

$E = 1\ V/m.$

$\varepsilon_r = 4.$

Average power density $P_{av} = \frac{E}{2\eta}$

$\eta = \sqrt{\frac{\mu}{\varepsilon}}$

$\varepsilon = \frac{\varepsilon_r}{4}\varepsilon_0$

$\eta = \sqrt{\frac{4\pi \times 10^{-7}}{8.854 \times 10^{-12} \times 4}}$

$\eta = 188.319$

$P_{av} = \frac{1}{2 \times 188.319}$

$P_{av} = 2.655\ mw/m^2.$

11) Find the minimum operating frequency of a rectangular waveguide, whose internal dimensions are $a = 4cm$, $b = 3cm$ in TE11 mode.

Given data: $a = 4cm$, $b = 3cm$

TE11 mode $\Rightarrow m = 1, n = 1$

$\frac{1}{\sqrt{\mu\varepsilon}} = 3 \times 10^8$

$f_c = \frac{1}{2\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$

$= \frac{3 \times 10^8}{2} \sqrt{\left(\frac{1}{4 \times 10^{-2}}\right)^2 + \left(\frac{1}{3 \times 10^{-2}}\right)^2}$

$f_c = 6.25\ GHz.$

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12) In a medium characterized by $\sigma = 0$, $\mu = \mu_0$, $\varepsilon = \varepsilon_0$ and $\epsilon = 20 \sin(10^8 t - \beta z)\, a_y$ V/m, calculate f_c for TE10 mode with a = 2 cm & b = 1 cm.

$$f_c = \frac{1}{2\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$\omega = 10^8$$

$$2\pi f = 10^8$$

$$f = \frac{10^8}{2\pi}$$

$$f = 15.92 \times 10^6 \text{ Hz}.$$

$$f_c = \frac{1}{2\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{1}{1 \times 10^{-2}}\right)^2 + 0} = 1.5 \times 10^{10} \text{ Hz}.$$

13) Give the relation between guide wavelength (λ_g) cutoff wavelength (λ_c) and free space wavelength λ_0 of a waveguide.

Guide wavelength $\lambda_g = \dfrac{\lambda_0}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$ or $\lambda = \dfrac{\lambda'}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$

$$\frac{1}{\lambda_0} \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = \frac{1}{\lambda_g}$$

$$\frac{1}{\lambda_0^2} - \frac{1}{\lambda_g^2} = \frac{1}{\lambda_c^2}$$

$\lambda_g = \lambda$ le
$\lambda_0 = \lambda'$

or

$$\frac{1}{\lambda'^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda_c^2}$$

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III unit

Problems

(1) A lossy dielectric has an intrinsic impedance of $200\angle 30^\circ \Omega$ at a particular radian frequency ω. If at that frequency, the plane wave propagating thro' the dielectric has the magnetic field component

$$H = 10 e^{-\alpha x} \cos\left(\omega t - \frac{1}{2} x\right) a_y \ A/m.$$

Find E & α. Determine skin depth and wave polarization.

(2) A uniform plane propagating in a medium has

$$E = 2 e^{-\alpha z} \sin\left(10^8 t - \beta z\right) a_y \ V/m$$

If the medium is characterized by $\varepsilon_r = 1$, $\mu_r = 20$, $\sigma = 3$ S/m

Find α, β and H.

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In a rectangular waveguide for wh
$a = 1.5\,cm$, $b = 0.8\,cm$, $\sigma = 0$, $\mu = \mu_0$, $\varepsilon = \varepsilon_0$
$\beta = 1$. The waveguide is operating in TM_{13} mode.

$$H_x = 2 \sin(\pi x/a) \cos(3\pi y/b) \sin(\pi \cdot 10^{11} t - \beta z)$$

Determine

(i) Cut-off frequency

(ii) phase constant

(iii) propagation constant

A std. air filled rectangular waveg
$a = 8.636\,cm$, $b = 4.318\,cm$ is fed by 4 GH
carrier from a co-axial cable.
Determine, whether TE_{10} mode will be propagated. If so, calculate the phase velocity and group velocity

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5) A rectangular waveguide has internal dimensions 2cm x 1cm and is propagating a 10GHz signal in the dominant mode (TE_{10}). Find the guided wavelength and wave impe

6) A rectangular metal wire, TE guide filled with a dielectric material of relative permitivity $\varepsilon_r = 4$ has the inside dimen 3.0cm x 1.2cm. Find the cut-off frequency for the dominant mode.

7) An air-filled rectangular waveguide, has inner dimensions of 3cm x 2cm. Find the wave impedance of the TE_{20} mode of propagati in the waveguide at a frequency of 30GHz. ($\eta_0 = 377\Omega$)

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A Co-axial cable with an inner diameter of 1mm and outer diameter of 2.4mm is filled with a dielectric of relative permitivity 10.89.

Given $\mu_0 = 4\pi \times 10^{-7}$ H/m

$$\varepsilon_0 = \frac{10^{-9}}{36\pi} \text{ F/m}$$

Find the characteristic impedance.

The electric field component of a time harmonic plane EM wave travelling in a non-magnetic lossless dielectric medium has an amplitude of 1V/m. If the relative permitivity of the medium is 4. Find the magnitude of the time average power density vector in (W/m^2).

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9.) Find the minimum Operating frequency of a rectangular waveguide, whose inter dimensions are $a = 4cm$, $b = cm$ in TE_{11}

11) In a medium, characterized by $\sigma = 0$, $\mu = \mu_0$, ε_0. and $E = 20 \sin(10^8 t - \beta z) a_y$ calculate β.

2) Give the relation between guide wavele (dg) cut-off wavelength (dc) and free Space wavelength (do) of a waveguide

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